

Sri Venkateswara College of Engineering, Sriperumbudur

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Electrical and Electronics Engineering

EE22811 Project Work – First Review

Project Type: In-house

Batch No.:15

Name of the Internal Guide: Mr. N. Suresh M.E., (Ph.D)

Name of the students with Roll No.

(1) Sanjay Kumar V - 2127220601063

(2) Shrimathi G - 2127220601070

Technical Area of the Project: Power Systems & Machine Learning

Title of the Project: Power Quality Disturbance Classification using Machine Learning

Abstract:

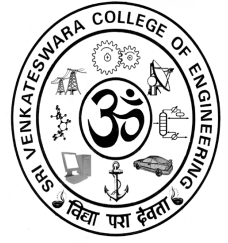
Power quality (PQ) disturbances such as voltage sag, swell, harmonics, and transients negatively impact the reliability and efficiency of modern power systems. Conventional monitoring methods based on FFT and RMS analysis often suffer from low accuracy and delayed response, limiting their effectiveness in real-time applications. Additionally, deep learning-based approaches require high computational resources and cloud dependency, leading to increased latency and cost.

This project proposes an intelligent edge-based power quality disturbance classification system using machine learning. Voltage and current signals are acquired using ZMPT101B and ACS712 sensors with ESP32 and Raspberry Pi 4 platforms at a 10 kHz sampling rate. Extracted features are processed using a Random Forest classifier optimized for embedded systems. The system classifies six major PQ disturbances with over 98% accuracy and inference time below 100 ms. RS485 communication and a Node-RED dashboard enable reliable monitoring and control. The proposed approach enhances grid stability, reduces energy loss, and supports sustainable smart grid development.

Signature of Internal Guide

Signature of the Students

Sri Venkateswara College of Engineering, Sriperumbudur
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AY 2025-26 EE22811 Project Work Progress Sheet
(21/01/2026 to 10/02/2026)



Batch No:15

Section: 'B'

Students Name & Roll No:

(1) Sanjay Kumar V - 2127220601063 (2) Shrimathi G - 2127220601070

Title of the project:

The above listed students have reported to me on _____ to finalize the project abstract and to review the power point slides presentation & simulation results. I hereby approve the same.

Date (project)	Abs Roll	Work done	Guide Approval with remarks
21/01/2026		Conducted an in-depth study on power quality disturbances such as voltage sag, swell, harmonics, interruptions, and transients in smart grid	
22/01/2026		Performed detailed literature survey on AI-based power quality monitoring and disturbance classification techniques.	
23/01/2026		Identified and finalized key electrical parameters including voltage variation, current harmonics, THD, frequency deviation, and waveform	
28/01/2026		Studied the working principles and performance characteristics of voltage and current sensors (ZMPT101B, ACS712) for PQ monitoring	
29/01/2026		Analyzed high-speed data acquisition methods and sensor interfacing techniques for 10 kHz signal sampling.	
30/01/2026		Investigated embedded platforms (ESP32, Raspberry Pi 4) for real-time signal processing and edge AI implementation.	
04/02/2026		Studied industrial communication protocols (RS485) and IoT-based monitoring frameworks for reliable data transmission and visualization.	
05/02/2026		Designed preliminary system architecture and block diagram for sensor integration, feature extraction, and classification workflow.	
06/02/2026		Initiated development planning for Node-RED-based dashboard for real-time PQ monitoring and performance analysis.	
09/02/2026		Reviewed project progress and model feasibility with project guide and incorporated suggested improvements in system design.	

Date:

Place:

Signature of Project Guide

